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### Integrated showerhead

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#### Abstract

A showerhead for a processing chamber includes a faceplate with a plurality of openings. A plurality of compartments are recessed into a top surface of the faceplate. The showerhead includes a plurality of MEMS devices. Each MEMS device is disposed in a corresponding compartment of the plurality of compartments. A printed circuit board including a plurality of ports therethrough is coupled to each MEMS device. Each MEMS device is configured to regulate a gas flow into each corresponding compartment through a corresponding port of the plurality of ports in the printed circuit board.

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name          | U.S. Cl.   | CPC              |
|--------------|-------------|------------------------|------------|------------------|
| 5853484      | 12/1997     | Jeong                  | 118/712    | C23C 16/44       |
| 6279402      | 12/2000     | Fisher                 | N/A        | N/A              |
| 6705345      | 12/2003     | Bifano                 | 251/129.01 | F16K 99/0001     |
| 10130958     | 12/2017     | Tam et al.             | N/A        | N/A              |
| 10403476     | 12/2018     | Gregor et al.          | N/A        | N/A              |
| 10804079     | 12/2019     | Gregor et al.          | N/A        | N/A              |
| 10934621     | 12/2020     | Sung et al.            | N/A        | N/A              |
| 2003/0019580 | 12/2002     | Strang                 | N/A        | N/A              |
| 2004/0040502 | 12/2003     | Basceri                | 118/715    | C23C<br>16/45574 |
| 2004/0040503 | 12/2003     | Basceri                | 118/715    | C23C<br>16/45531 |
| 2013/0012029 | 12/2012     | Vermeer et al.         | N/A        | N/A              |
| 2020/0365386 | 12/2019     | Benjamin Raj<br>et al. | N/A        | N/A              |
| 2021/0096011 | 12/2020     | Smith et al.           | N/A        | N/A              |

### FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 200489691  | 12/2018          | KR      | N/A |
| 2019078919 | 12/2018          | WO      | N/A |

### OTHER PUBLICATIONS

International Search Report and Written Opinion dated May 4, 2023 for Application No. PCT/US2023/010578. cited by applicant

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## Background/Summary

## BACKGROUND

### Field

(1) Embodiments of the present disclosure generally relate to apparatus and methods utilized in the manufacture of semiconductor devices. More particularly, embodiments of the present disclosure relate to components, such as a showerhead, of a processing chamber for forming semiconductor devices.

### Description of the Related Art

(2) Semiconductor substrates are processed for a wide variety of applications, including the fabrication of integrated devices and microdevices. During processing, the substrate is positioned on a substrate support within a process chamber. The interior of the process chamber is placed under vacuum while the substrate is processed by exposure to process gases. Some processes involve etching material away from the substrate, and other processes involve the deposition of material onto the substrate. The uniformity of the etch or of the material deposited on the substrate may be affected by the distribution of process gases within the process chamber. In some process chambers, a showerhead distributes the process gas. The pattern and sizes of holes in the showerhead may be optimized for the distribution of a certain process gas for a particular processing operation, but may not be optimized for a different process gas or for distributing process gasses in a different processing operation.

(3) Thus, there is a need for improved process chambers that facilitate effective control over process gas distribution.

### SUMMARY

(4) The present disclosure generally relates to components, such as a showerhead, of a substrate processing chamber for forming semiconductor devices. In one embodiment, a showerhead for a processing chamber includes a faceplate. The faceplate includes a bottom surface, a top surface, and a plurality of openings extending from the top surface to the bottom surface. A printed circuit board is coupled to the faceplate. The showerhead further includes a plurality of MEMS devices coupled to the printed circuit board, each MEMS device associated with one or more unique openings of the plurality of openings, and configured to regulate a gas flow through the corresponding one or more unique openings. The showerhead further includes a plurality of local controllers coupled to the printed circuit board, each local controller configured to control operation of a corresponding MEMS device of the plurality of MEMS devices independently of an operation of other MEMS devices of the plurality of MEMS devices.

(5) In another embodiment, a showerhead for a processing chamber includes a printed circuit board including a plurality of ports therethrough. The showerhead further includes a faceplate. The faceplate includes a plurality of MEMS modules coupled to the printed circuit board. Each MEMS module includes: a body; sidewalls extending below the body to a base, the base including one or more holes; and a MEMS device operable to control gas flow through at least one of the plurality of ports.

(6) In another embodiment, a processing chamber includes a chamber body and a showerhead disposed in the chamber body. The faceplate includes a bottom surface, a top surface, a plurality of compartments recessed into the top surface, and a plurality of openings extending from each compartment to the bottom surface. The showerhead further includes a plurality of MEMS devices, each MEMS device in a corresponding compartment of the plurality of compartments, and configured to regulate a gas flow into each corresponding compartment. The showerhead further includes a printed circuit board coupled to the top surface of the faceplate and to each MEMS device. The showerhead further includes a controller coupled to the printed circuit board, and configured to control operations of at least one MEMS device of the plurality of MEMS devices independently of operations of other MEMS devices of the plurality of MEMS devices.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, as the disclosure may admit to other equally effective embodiments.

(2) FIG. 1 is a schematic cross-sectional view of a processing chamber.

(3) FIG. 2A is an isometric view of the top of an exemplary MEMS device.

(4) FIG. 2B is an isometric view of the bottom of the MEMS device of FIG. 2A.

(5) FIGS. 2C and 2D are plan views of the top of the MEMS device of FIG. 2A.

(6) FIG. 2E is a plan view of the top of another exemplary MEMS device.

(7) FIG. 3 is a schematic cross-sectional side view of an exemplary showerhead.

(8) FIGS. 4A and 4B are schematic cross-sectional side view of exemplary showerheads.

(9) FIGS. 5A and 5B are schematic cross-sectional side views of exemplary showerheads.

(10) FIGS. 6A-6G are schematic cross-sectional side views of configurations of MEMS modules.

(11) FIG. 7 is a schematic plan view of a faceplate of a showerhead.

(12) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in one or more other embodiments without further recitation.

### DETAILED DESCRIPTION

(13) The present disclosure concerns components, such as a showerhead, of a processing chamber for forming semiconductor devices. Embodiments of the present disclosure provide showerheads that can be readily configured for use with any one or more of a plurality of gases used in the processing of substrates. Example gases include silicon-containing gases, oxygen-containing gases, nitrogen-containing gases, hydrogen-containing gases, argon-containing gases, and metal-containing gases.

(14) FIG. 1 illustrates a schematic cross-sectional view of a processing chamber **100**. As illustrated, the processing chamber **100** is configured as a Plasma Enhanced Chemical Vapor Deposition (PECVD) chamber, although in some embodiments, processing chamber **100** may be configured to perform another plasma-enhanced processing operation (such as etching or physical vapor deposition) or a processing operation that does not involve plasma (such as chemical vapor deposition). The processing chamber **100** features a chamber body **102**, a substrate support **104** disposed inside the chamber body **102**, and a lid **106** coupled to the chamber body **102**, and enclosing the substrate support **104** in a processing volume **120**. The substrate support **104** is configured to support a substrate **154** thereon during processing. The substrate **154** is provided to the processing volume **120** through an opening **126**. While the embodiment of FIG. 1 is directed to a PECVD chamber, the lid **106** and substrate support **104** of FIG. 1 may be used with other processing chambers that utilize plasma generated in the processing volume **120**. Additionally, the lid **106** and substrate support **104** of FIG. 1 may be used with other processing chambers that do not utilize plasma generated in the processing volume **120**.

(15) As illustrated, a showerhead **112** contains or serves as an electrode, and is coupled to a power source **128** through a match circuit. The power source **128** is a radio frequency (RF) power source that is electrically coupled to the electrode **108**. Further, the power source **128** provides between about 100 Watts and about 3,000 Watts at a frequency of about 50 kHz to about 13.6 MHz. In some embodiments, the power source **128** can be pulsed during various operations. The electrode **108**

and power source **128** facilitate control of a plasma formed within the processing volume **120**.

(16) The substrate support **104** contains, or is formed from, one or more metallic or ceramic materials. Exemplary metallic or ceramic materials include one or more metals, metal oxides, metal nitrides, metal oxynitrides, or any combination thereof. For example, the substrate support **104** may contain or be formed from aluminum, aluminum oxide, aluminum nitride, aluminum oxynitride, or any combination thereof.

(17) As illustrated, an electrode **122** is embedded within the substrate support **104**, but alternatively may be coupled to a surface of the substrate support **104**. The electrode **122** is coupled to a power source **136**. It is contemplated that the power source **136** may be DC power, pulsed DC power, radio frequency (RF) power, pulsed RF power, or any combination thereof. The power source **136** is configured to drive the electrode **122** with a drive signal to generate a plasma within the processing volume **120**. It is contemplated that the drive signal may be one of a DC signal and a varying voltage signal (e.g., RF signal). Further, the electrode **122** may alternatively be coupled to the power source **128** instead of the power source **136**, and the power source **136** may be omitted.

(18) Plasma is generated in the processing volume **120** via the power source **128** and the power source **136**. An RF field is created by driving at least one of the showerhead **112** electrode and the electrode **122** with drive signals to facilitate the formation of a plasma within the processing volume **120**. The presence of a plasma facilitates processing of the substrate **154**, for example, the deposition of a film onto a surface of the substrate **154** or the etching of material from a surface of the substrate **154**.

(19) An exhaust port **156** is coupled to a vacuum pump **157**. The vacuum pump **157** removes excess process gases or by-products from the processing volume **120** via the exhaust port **156** during and/or after processing.

(20) A gas supply source **111** includes one or more gas sources. The gas supply source **111** is configured to deliver the one or more gases from the one or more gas sources through the showerhead **112** and into the processing volume **120**. Each of the one or more gas sources provides a process gas such as silane, disilane, tetraethyl orthosilicate (TEOS), germane, a metal halide (such as titanium tetrachloride, tantalum pentachloride, tungsten hexafluoride), an organometallic (such as tetrakis(dimethylamido) titanium, pentakis(dimethylamido) tantalum), ammonia, oxygen (O.sub.2), hydrogen peroxide, hydrogen, diborane, chlorine (Cl.sub.2), sulfur hexafluoride, a hydrocarbon (generically C.sub.xH.sub.y), among others. In some embodiments, the process gas may be ionized to form a plasma within the processing volume **120**. For example, one or more of a carrier gas and an ionizable process gas are provided into the processing volume **120** to process the substrate **154**. When processing a 300 mm substrate, the process gases are introduced to the processing chamber **100** at a flow rate from about 6500 sccm to about 8000 sccm, from about 100 sccm to about 10,000 sccm, or from about 100 sccm to about 1000 sccm. Alternatively, other flow rates may be utilized. In some examples, a remote plasma source can be used to deliver plasma to the processing chamber **100** and can be coupled to the gas supply source **111**.

(21) The showerhead **112** features openings **118** for admitting a process gas or gases into the processing volume **120** from the gas supply source **111**. The process gases are supplied to the processing chamber **100** via the gas feed **114**, and the process gases enter a plenum **116** prior to flowing through the openings **118**. In some embodiments, different process gases that are flowed simultaneously during a processing operation enter the processing chamber **100** via separate gas feeds and separate plenums prior to entering the processing volume **120** through the showerhead **112**.

(22) Gas flow through the openings **118** of the showerhead **112** is regulated by one or more micro-electro-mechanical systems (MEMS devices) disposed in the showerhead **112**. In some embodiments, it is contemplated that gas flow through an individual opening **118** and/or through a cluster of openings **118** may be regulated by a MEMS device. In some embodiments, it is contemplated that gas flow through all openings **118** is regulated by a plurality of MEMS devices.

In an example, each MEMS device regulates gas flow through one or more openings **118** such that gas flow through any single opening **118** is regulated by a corresponding MEMS device. It is contemplated that the regulation of gas flow by a MEMS device includes permitting a maximum flow of gas through an individual opening **118** and/or through a cluster of openings **118**. It is contemplated that the regulation of gas flow by a MEMS device includes preventing any flow of gas through an individual opening **118** and/or through a cluster of openings **118**. It is contemplated that the regulation of gas flow by a MEMS device includes controlling a flow of gas through an individual opening **118** and/or through a cluster of openings **118** such that the flow is greater than zero and less than a maximum flow of gas through the individual opening **118** and/or through the cluster of openings **118**.

(23) FIGS. 2A-2D illustrate an exemplary MEMS device **200**. FIG. 2A is an isometric view of the top of the MEMS device **200**, and FIG. 2B is an isometric view of the bottom of the MEMS device **200**. The MEMS device **200** includes a body **202**, having an orifice **204** therethrough. Although illustrated as substantially rectangular, it is contemplated that the orifice **204** may have any appropriate cross-sectional shape, such as circular, elliptical, triangular, and the like. Furthermore, in some embodiments, the orifice **204** may include multiple orifices. Moreover, it is contemplated that one variant of MEMS device **200** may have an orifice **204** that has a cross-sectional size different from the cross-sectional size of the orifice **204** of another variant of MEMS device **200**.

(24) A skirt **206** extends from the body **202** at the bottom of the MEMS device **200**. A valve member **210** is mounted on the body **202** and regulates fluid flow through the orifice **204**. The valve member **210** is electrically conductive. In some embodiments, the valve member **210** is metallic. The valve member **210** passes an electrical current that flows between contacts **230**, **231**, to which the valve member **210** is connected. As illustrated, in some embodiments, the MEMS device **200** includes contacts **232**, **233** that are configured for connection to a heater, such as a wire through which an electrical current is passed in order to induce heating. In some of such embodiments, the heater is integrated with the MEMS device **200**. Alternatively, the heater may be a separate component configured to be plugged into the MEMS device **200**. In some embodiments, the heater may be omitted.

(25) As illustrated, in some embodiments, the MEMS device **200** includes contacts **234**, **235** that are configured for connection to a sensor **236**. In some of such embodiments, the sensor **236** is integrated with the MEMS device **200**. Alternatively, the sensor **236** may be a separate component configured to be plugged into the MEMS device **200**. It is contemplated that the sensor **236** may be configured to measure one or more of pressure, temperature, or flow rate. In an example, measurements of the flow rate of a fluid through the orifice **204** may be derived at least in part from measurements of pressure obtained from the sensor **236**. In some embodiments, the sensor **236** may be omitted.

(26) FIG. 2C is a plan view of the top of the MEMS device **200**. The valve member **210** of the MEMS device **200** includes a first section **212** and a second section **214**. At an end **216** of the valve member **210** near the orifice **204**, the first and second sections **212**, **214** are connected together. At an opposite end **218** of the valve member **210**, the first section **212** is connected to contact **230**, and the second section **214** is connected to contact **231**, but the first **212** and second **214** sections are not connected together. The first section **212** is nominally thicker than the second section **214**, but includes a void **220**. The void **220** is illustrated as two connected rectangles, however, it is contemplated that the void **220** may have or include any suitable shape, such as one or more triangles, one or more squares, one or more circles, one or more ellipses, or one or more of any other shape. The void **220** divides the first section into a relatively thick portion **222** and one or more relatively thin portions **224**. An end portion **226** of the first section **212** is relatively thick, and is positioned at the end **216** near to the orifice **204**. In the configuration illustrated, the orifice **204** is at least partially uncovered by the end portion **226**, thereby permitting gas to flow through the orifice **204**. The end portion **226** is configured to at least partially obscure the orifice **204** during

operation of the MEMS device **200**.

(27) FIG. 2D is a plan view of the top of the MEMS device **200** when an electrical current is passed through the valve member **210**. Because the first section **212** and second section **214** are connected together at end **216** but not connected together at end **218**, a voltage applied at the contacts **230**, **231** causes a current to flow through the first section **212** and through the second section **214**. The thicknesses of the first **212** and second **214** sections impact the electrical resistances of the first and second sections **212**, **214**—the thicker the section, the lower the resistance. When a current flows through the valve member **210**, the second section **214** and the relatively thin portions **224** of the first section **212** experience greater heating than the relatively thick portion(s) **222** of the first section **212**.

(28) Because of the difference in heating, the relatively thick portion(s) **222** of the first section **212** do not experience as much thermal expansion as do the second section **214** and the relatively thin portions **224** of the first section **212**. Thus, the first section **212** does not linearly elongate to the same extent as does the second section **214**. Because the first section **212** and the second section **214** are connected together at end **216**, elongation of the second section **214** causes the first section **212** to deform into an “S” shape, facilitated by the void **220**. The end **216** deflects in the direction of arrow **228**, thus causing the end portion **226** of the first section **212** to at least partially obscure the orifice **204**. In some embodiments, it is contemplated that the end **216** deflects to such an extent that the end portion **226** completely obscures the orifice **204**. In some of such embodiments, the end portion **226** completely blocks passage of gas through the orifice **204**.

(29) As illustrated in FIG. 2D, in some embodiments, elongation of the valve member **210** due to heating is compensated by deformation of the first section **212** into the “S” shape. In this way, the orientation of the end portion **226** is maintained during travel, and the end portion **226** can completely obscure the orifice **204**. However, in some embodiments, the end portion **226** does not completely obscure the orifice **204**, but only partially obscures the orifice **204** when an electrical current is applied to the valve member **210**.

(30) When the current passing through the valve member **210** is reduced to a smaller magnitude, or is completely ceased, the valve member **210** cools down, experiences thermal contraction, and returns towards the position illustrated in FIG. 2C. Thus, the positioning of the end portion **226** with respect to the orifice **204** is controlled by modifying the current passing through the valve member **210**. Consequently, the amount of fluid flow through the orifice **204** is controlled by adjusting the magnitude of the current passing through the valve member **210**. In an example, the orifice **204** is completely uncovered at zero current through the valve member **210**, the orifice **204** is completely covered by the end portion **226** at a prescribed maximum current through the valve member **210**, and the orifice **204** is partially covered by the end portion **226** at a given fraction of the prescribed maximum current through the valve member **210**. In such an example, a maximum fluid flowrate through the orifice **204** is realized at zero current through the valve member **210**, zero flow through the orifice **204** is realized at the prescribed maximum current through the valve member **210**, and a fraction of the maximum fluid flowrate through the orifice **204** is realized at the given fraction of the prescribed maximum current through the valve member **210**.

(31) In some embodiments, the applied current through the valve member **210** may be adjusted in steps, thereby providing one or more intermediate positions of the end portion **226** of the valve member **210** between fully uncovering the orifice **204** and completely obscuring the orifice **204**. In such embodiments, the MEMS device **200** provides an intermediate fluid flowrate between zero flow and the maximum flowrate corresponding to each intermediate position of the end portion **226**. In an example, the resulting fluid flowrate through the orifice **204** may be varied in increments (such as in 5% increments, 10% increments, or 20% increments) from zero to the maximum flow.

(32) In some embodiments, the applied current through the valve member **210** may be continuously variable, thereby providing a continuously variable position of the end portion **226** of the valve member **210** between fully uncovering the orifice **204** and completely obscuring the orifice **204**. In

such embodiments, the MEMS device **200** provides a continuously variable fluid flowrate through the orifice **204** between zero flow and the maximum flowrate, the resulting flowrate corresponding to the intermediate position of the end portion **226**.

(33) In some embodiments, the applied current through the valve member **210** may be adjusted stepwise over a portion of the range from zero to maximum current, and may be continuously variable over another portion of the range from zero to maximum current. In such embodiments, the resulting fluid flowrate through the orifice **204** may be varied in steps over a portion of the range from zero to maximum flow, and may be continuously variable over another portion of the range from zero to maximum flow. In an example, the resulting fluid flowrate through the orifice **204** may be varied in steps from zero to 20% of the maximum flow, and may be varied continuously from 20% to the maximum flow.

(34) Although FIGS. 2A-2D illustrate the MEMS device **200** to be configured with the orifice **204** normally open with no current applied, in some embodiments, the MEMS device **200** may be configured with the orifice **204** normally closed with no current applied. In such embodiments, the starting position for the valve member **210** includes the end portion **226** obscuring the orifice **204**. In an example, the end portion **226** completely blocks passage of gas through the orifice **204**. The application of a current through the valve member **210** causes deflection of the end **216** of the valve member, moving the end portion **226** to at least partially uncover the orifice **204**, thereby permitting gas to flow through the orifice **204**.

(35) Any of the arrangements of MEMS devices in the present disclosure may include MEMS devices configured with a normally open orifice. Any of the arrangements of MEMS devices in the present disclosure may include MEMS devices configured with a normally closed orifice. Any of the arrangements of MEMS devices in the present disclosure may include a combination of MEMS devices configured with a normally open orifice and MEMS devices configured with a normally closed orifice.

(36) FIG. 2E is a plan view of the top of an exemplary MEMS device **250**. MEMS device **250** is similar to MEMS device **200**, but includes two orifices **254A**, **254B** and two valve members **260A**, **260B**. Valve member **260A** is connected to contacts **280A**, **281A**, and regulates fluid flow through orifice **254A**. Valve member **260B** is connected to contacts **280B**, **281B**, and regulates fluid flow through orifice **254B**. In some embodiments, a dedicated heater is associated with each orifice **254A**, **254B**; the heater associated with orifice **254A** is connected to contacts **282A**, **283A**, and the heater associated with orifice **254B** is connected to contacts **282B**, **283B**. A sensor (**236**, FIG. 2B), as described above, is connected to contacts **284**, **285**. In some embodiments, it is contemplated that valve member **260A** and valve member **260B** may be operated independently, and therefore orifice **254A** and orifice **254B** may be suitable for coupling to separate gas supplies.

(37) To inhibit corrosion and/or reduce a probability that a valve member of a MEMS device may stick in position and become inoperable, it is contemplated that surfaces of each component of MEMS devices **200**, **250** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al.sub.2O.sub.3), yttrium oxides (such as Y.sub.2O.sub.3), silicon oxides (such as SiO.sub.x), titanium oxides (such as TiO.sub.2), and the like.

(38) FIG. 3 is a schematic cross-sectional side view of an exemplary showerhead **300**. It is contemplated that the configuration of showerhead **300** may be used as showerhead **112** of FIG. 1. Showerhead **300** includes a faceplate **310** with openings **318**, through which gases flow from the plenum **116** into the processing volume (**120**, FIG. 1) of a processing chamber, such as processing chamber **100**. A top surface **312** of the faceplate **310** includes compartments **314**. As illustrated, each compartment **314** is recessed into the top surface **312**. In some embodiments, the compartments **314** may not be recessed into the top surface **312**. A cluster of openings **318** is associated with each compartment **314**.



(39) A MEMS device **320** is associated with each compartment **314**. As illustrated, in embodiments in which a compartment **314** is recessed into the top surface **312** of the faceplate **310**, the MEMS device **320** may be at least partially disposed in a corresponding compartment **314**. It is contemplated that the MEMS device **320** may be configured similarly to MEMS device **200** or MEMS device **250**. Each MEMS device **320** is schematically depicted to include an orifice **322**, a valve member **324**, a heater **326**, and a sensor **328**, such as described above for MEMS device **200**. Each MEMS device **320** is coupled to a printed circuit board (PCB) **330**. In some embodiments, each MEMS device **320** is soldered to the PCB **330**. In some of such embodiments, the solder surrounds the orifice **322** and provides a seal between the PCB **330** and each MEMS device **320**. Each contact of each MEMS device **320** is connected to the PCB **330**. The sensor **328**, the heater **326**, and the valve member **324** of each MEMS device **320** receive electrical power via the PCB **330**. The PCB **330** is coupled to a master controller **350** for the transmission of power and/or control signals and/or telemetry with each MEMS device **320**.

(40) The PCB **330** includes a port **332** associated with each MEMS device **320**. When the valve member **324** of a MEMS device **320** permits gas to flow through the corresponding orifice **322**, gas in the plenum **116** can flow through the corresponding port **332** in the PCB **330** and through the orifice **322** into the corresponding compartment **314** of the faceplate **310**. In some embodiments, the gas is heated by the heater **326**. The gas flows from the compartment **314** through the corresponding openings **318** in the faceplate **310** into the processing volume (**120**, FIG. 1) of the processing chamber.

(41) As illustrated, in some embodiments, showerhead **300** may include one or more local controllers **329**. Each local controller **329** may be associated with, and programmed to control, a corresponding single MEMS device **320** or a corresponding group of MEMS devices **320**. In an example, each local controller **329** includes an application-specific integrated circuit (ASIC). In some embodiments, each local controller **329** may be integrated into a MEMS device **320**. As illustrated, in some embodiments, each local controller **329** may be coupled to the PCB **330** separate from the MEMS device **320**. In some embodiments, the local controller **329** includes an electromagnetic shield. To inhibit corrosion, it is contemplated that surfaces of the local controller **329** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al.sub.2O.sub.3), yttrium oxides (such as Y.sub.2O.sub.3), silicon oxides (such as SiO.sub.x), titanium oxides (such as TiO.sub.2), and the like.

(42) In some embodiments, the local controller **329** receives commands from master controller **350** via the PCB **330**. It is contemplated that the commands may be in the form of a signal that is addressed to correspond with a specific device, such as a specific MEMS device **320**. Each local controller **329** is programmed to recognize command signals addressed to correspond with devices under the purview of local controller **329**, and controls the devices according to the commands received. In some embodiments, each local controller **329** is programmed to ignore command signals that are not addressed to correspond with any of the devices under the purview of local controller **329**.

(43) In some embodiments, each MEMS device **320** is independently addressable via a corresponding local controller **329**, such that the operation of each MEMS device **320** can be controlled without changing the operating status of any other MEMS device **320**. In some embodiments, each MEMS device **320** is assigned to one or more groups of MEMS devices **320**, and each group of MEMS devices **320** is independently addressable via one or more corresponding local controllers **329**. In such embodiments, the operation of each MEMS device **320** within a defined group can be controlled without changing the operating status of any other MEMS device **320** that is not within the defined group.

(44) In an example, each MEMS device **320** or group of MEMS devices **320** are associated with a discrete zone of the faceplate **310**, such as illustrated by any zone **710** of faceplate **700** in FIG. 7.

The control of each MEMS device **320**, or group of MEMS devices **320**, independently of other MEMS devices of showerhead **300** facilitates the adjustment of gas flow distribution across the zones of the faceplate **310**.

(45) In an example, a cluster of MEMS devices **320** at the center of the faceplate **310** are assigned to “Group A” and a cluster of MEMS devices **320** at an edge of the faceplate **310** are assigned to “Group B.” The MEMS devices **320** of Group A can be controlled independently from the MEMS devices of Group B. Additionally, the MEMS devices **320** of Group A can be controlled via a command addressed to the group, and the MEMS devices **320** of Group B do not respond to the command addressed to Group A. In such an example, the MEMS devices **320** of Groups A and B can be controlled to adjust the quantity of process gas being delivered to the center of a substrate, such as substrate **154**, relative to the quantity of process gas being delivered to the edge of the substrate.

(46) In another example, a processing chamber, such as processing chamber **100**, has an exhaust port (**156**, FIG. **1**) located in an off-center position, which causes variations in gas flow at different locations within the processing volume (**120**, FIG. **1**) of the processing chamber. Such azimuthal variations in gas flow can result in uneven processing of a substrate, such as a disparity of film thickness across the substrate. In such an example, a cluster of MEMS devices **320** near the exhaust port are assigned to “Group C” and a cluster of MEMS devices **320** further away from the exhaust port are assigned to “Group D.” The MEMS devices **320** of Group C can be controlled independently from the MEMS devices of Group D. Additionally, the MEMS devices **320** of Group C can be controlled via a command addressed to the group, and the MEMS devices **320** of Group D do not respond to the command addressed to Group C. In such an example, the MEMS devices **320** of Groups C and D can be controlled to adjust the quantity of process gas being delivered to a portion of the substrate near to the exhaust port relative to the quantity of process gas being delivered to a portion of the substrate further away from the exhaust port.

(47) In some embodiments, one or more MEMS devices **320** may be controlled according to a hierarchy of commands such that MEMS devices **320** not within a specific hierarchical set of MEMS devices **320** are unaffected by operating commands addressed to MEMS devices **320** within the specific hierarchical set. In an example, a particular MEMS device **320** is allocated to a small group of MEMS devices (“Group E1”) which is part of a larger group of MEMS devices (“Group E”). In this example, the particular MEMS device **320** is also allocated to a different group of MEMS devices (“Group F”) that contains other MEMS devices that are not within Group E. The particular MEMS device **320** can be controlled by commands addressed only to that particular MEMS device **320**, and no other MEMS device will respond to those commands. The particular MEMS device **320** can be controlled also by commands addressed only to Group E1. All MEMS devices in Group E1, including that particular MEMS device **320**, will respond to those commands, but no other MEMS device will respond to those commands. The particular MEMS device **320** can be controlled also by commands addressed only to Group E. All MEMS devices in Group E, including that particular MEMS device **320**, will respond to those commands, but no other MEMS device will respond to those commands. The particular MEMS device **320** can be controlled also by commands addressed only to Group F. All MEMS devices in Group F, including that particular MEMS device **320**, will respond to those commands, but no other MEMS device—including MEMS devices within Group E1 or Group E, unless those other MEMS devices are also allocated to Group F—will respond to those commands.

(48) In embodiments in which local controller **329** is omitted, master controller **350** operates each MEMS device **320** via electrically conductive lines embedded in the PCB **330**.

(49) FIG. **4A** is a schematic cross-sectional side view of an exemplary showerhead **400A**. It is contemplated that the configuration of showerhead **400A** may be used as showerhead **112** of FIG. **1**. Showerhead **400A** includes a manifold **440** disposed on a PCB **430**. The manifold **440** includes a first conduit **442** for passage of a first gas, and a second conduit **444** for passage of a second gas. It

is contemplated that the first conduit **442** is isolated from the second conduit **444** such that the first gas and the second gas do not mix in the manifold **440**. First ducts **446** from the first conduit **442** are aligned with first ports **432** in the PCB **430**. Second ducts **448** from the second conduit **444** are aligned with second ports **434** in the PCB **430**. An interface **438** between the manifold **440** and the PCB **430** is sealed, such as by bonding the manifold **440** to the PCB **430**, to inhibit mixing of the first gas and the second gas at the interface **438**.

(50) In some embodiments, manifold **440** includes one or more additional conduits and corresponding ducts configured to convey one or more additional gases. In such embodiments, the one or more additional conduits may be isolated from the first conduit **442** and the second conduit **444**. Furthermore, it is contemplated that the PCB **430** may include additional ports aligned with the additional ducts.

(51) The showerhead **400A** includes a faceplate **410** with openings **418**, through which gases flow into the processing volume (**120**, FIG. **1**) of a processing chamber, such as processing chamber **100**. As illustrated, in some embodiments, the manifold **440** and PCB **430** are coupled to the faceplate **410** by one or more fastener **416**, such as a screw or a bolt. A top surface **412** of the faceplate **410** includes compartments **414**. As illustrated, each compartment **414** is recessed into the top surface **412**. In some embodiments, the compartments **414** may not be recessed into the top surface **412**. A cluster of openings **418** is associated with each compartment **414**.

(52) A spacer **460** is associated with each compartment **414**. As illustrated, in embodiments in which a compartment **414** is recessed into the top surface **412** of the faceplate **410**, the spacer **460** may be at least partially disposed in a corresponding compartment **414**. Each spacer **460** includes sidewalls **462** and a floor **464**. Although the sidewalls **462** are illustrated as extending to form a shroud **466** below the floor **464**, in some embodiments, the shroud **466** may be omitted. Holes **468** in the floor **464** facilitate communication of gas to the openings **418** of the faceplate **410**. As illustrated, in some embodiments, a diffuser **470** is disposed above the holes **468**. The diffuser **470** can promote a uniform distribution of gas through the holes **468**. In some embodiments, the diffuser **470** filters out particles entrained in the gas. Example diffusers **470** include a mesh (such as a sintered mesh), a porous metal filter, a foam (such as porous PTFE foam), or the like. In some embodiments, the diffuser **470** may be omitted.

(53) The sidewalls **462** of the spacer **460** extend above the floor **464** to the PCB **430**. As illustrated, in some embodiments, a gasket **472** seals an interface between the spacer **460** and the PCB **430**. The gasket **472** may be made of any suitable material that can form a pressure seal and is resistant to chemical attack, such as an elastomer/thermoplastic material (such as an FKM type material, such as polyvinylidene difluoride (PVDF), including PVDF in the form of a closed cell foam), or the like. Each spacer **460** encloses a void space **474** between the PCB **430** and each corresponding compartment **414**. The ports **432**, **434** of the PCB **430** convey gas into the void spaces **474** enclosed by each spacer **460**.

(54) To inhibit corrosion, it is contemplated that the spacer **460** may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal such as titanium. Additionally, or alternatively, surfaces of the spacer **460** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as  $\text{Al}_2\text{O}_3$ ), yttrium oxides (such as  $\text{Y}_2\text{O}_3$ ), silicon oxides (such as  $\text{SiO}_x$ ), titanium oxides (such as  $\text{TiO}_2$ ), and the like.

(55) It is contemplated that clusters of MEMS devices of showerhead **400A** may be controlled in similar ways to the examples provided above with respect to clusters of MEMS devices **320** of showerhead **300**.

(56) FIG. **4A** illustrates three exemplary configurations in which one or more MEMS devices **420**, **421**, **422**, **423** with spacers **460**, and attached to the PCB **430**, can be associated with each compartment **414** of the faceplate **410**. Each pairing of one or more MEMS devices **420**, **421**, **422**,

**423** with a corresponding compartment **414** may be considered as a discrete unit; each unit is described below.

(57) In a first unit **482**, the PCB **430** includes a port **432** aligned with a first duct **446** of the manifold **440**, but does not include a port corresponding to a second duct **448** of the manifold **440**. Consequently, the first unit **482** is configured to manage gas supplied via the first conduit **442** of the manifold **440**, but is not configured to manage gas supplied via the second conduit **444** of the manifold **440**. However, in alternative embodiments, the first unit **482** may be configured to manage gas supplied via the second conduit **444** of the manifold **440**, but not configured to manage gas supplied via the first conduit **442** of the manifold **440**. In such embodiments, the PCB **430** includes a port aligned with a second duct **448** of the manifold **440**, but does not include a port corresponding to a first duct **446** of the manifold **440**.

(58) A MEMS device **420** within the void space **474** is coupled to the PCB **430**, and regulates gas flow through the port **432** of the PCB **430**. It is contemplated that the MEMS device **420** may be configured similarly to MEMS device **200**. MEMS device **420** includes an orifice **451** and a valve member **452**. In some embodiments, the MEMS device **420** is soldered to the PCB **430**. In some of such embodiments, the solder surrounds the orifice **451** and provides a seal between the PCB **430** and the MEMS device **420**. In some embodiments, the MEMS device **420** includes a sensor, such as sensor **236** (FIG. 2B). In some embodiments, a sensor separate from the MEMS device **420** is coupled to the PCB **430** within the void space **474**. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, the MEMS device **420** includes a heater. In some embodiments, as illustrated, a heater **424** separate from the MEMS device **420** is coupled to the PCB **430** within the void space **474**. The sensor, the heater, and the valve member associated with each MEMS device **420**, and heater **424** (if present) receive electrical power via the PCB **430**.

(59) As illustrated in the first unit **482**, in some embodiments, a local controller **429** may be associated with, and programmed to control, the MEMS device **420**. In an example, the local controller **429** includes an application-specific integrated circuit (ASIC). In some embodiments, the local controller **429** is integrated into the MEMS device **420**. In some embodiments, the local controller **429** is coupled to the PCB **430** separate from the MEMS device **420**. In some embodiments, the local controller **429** includes an electromagnetic shield. To inhibit corrosion, it is contemplated that surfaces of the local controller **429** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al.sub.2O.sub.3), yttrium oxides (such as Y.sub.2O.sub.3), silicon oxides (such as SiO.sub.x), titanium oxides (such as TiO.sub.2), and the like.

(60) In embodiments that include the local controller **429**, the local controller **429** controls operation of the MEMS device **420** and/or the heater **424**. In some embodiments, the local controller **429** receives commands from master controller **450** via the PCB **430**. It is contemplated that the commands may be in the form of a signal that is addressed to correspond with a specific device, such as MEMS device **420** or heater **424**. The local controller **429** in the first unit **482** is programmed to recognize command signals addressed to correspond with devices in the first unit **482** (such as MEMS device **420** or heater **424**), and controls the devices according to the commands received. In some embodiments, the local controller **429** is programmed to ignore command signals that are not addressed to correspond with any of the devices under the purview of local controller **429**, such as devices in the first unit **482**. In some embodiments, the local controller **429** may be programmed to control one or more devices that are not in the first unit **482**. In an example, the local controller **429** is programmed to control one or more MEMS devices in one or more units in addition to control MEMS device **420** and heater **424** of the first unit **482**.

(61) In some embodiments, each device in the first unit **482** is independently addressable via a corresponding local controller **429**, such that the operation of each device in the first unit **482** can be controlled without changing the operating status of any other device of showerhead **400A**. In

some embodiments, each device in the first unit **482** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers **429**. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **400A** that is not within the defined group.

(62) In embodiments in which the local controller **429** is omitted, master controller **450** operates the MEMS device **420** and the heater **424** via electrically conductive lines embedded in the PCB **430**.

(63) In a second unit **484**, the PCB **430** includes a first port **432** aligned with a first duct **446** of the manifold **440**, and a second port **434** aligned with a second duct **448** of the manifold **440**. Consequently, the second unit **484** is configured to manage gas supplied via the first conduit **442** of the manifold **440** and via the second conduit **444** of the manifold **440**.

(64) A first MEMS device **421** in the void space **474** of the second unit **484** is coupled to the PCB **430**, and regulates gas flow through the first port **432** of the PCB **430**. A second MEMS device **422** in the void space **474** of the second unit **484** is coupled to the PCB **430**, and regulates gas flow through the second port **434** of the PCB **430**. It is contemplated that each MEMS device **421**, **422** may be configured similarly to MEMS device **200**. MEMS device **421** includes an orifice **425** and a valve member **426**. MEMS device **422** includes an orifice **427** and a valve member **428**. In some embodiments, each MEMS device **421**, **422** is soldered to the PCB **430**. In some of such embodiments, the solder surrounds each orifice **425**, **427** and provides a seal between the PCB **430** and each MEMS device **421**, **422**. In some embodiments, at least one of the MEMS devices **421**, **422** includes a sensor, such as sensor **236** (FIG. 2B). In some embodiments, a sensor separate from the MEMS devices **421**, **422** is coupled to the PCB **430**. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, each MEMS device **421**, **422** includes a heater. In some embodiments, a heater (such as heater **424** of the first unit **482**) separate from the MEMS devices **421**, **422** is coupled to the PCB **430**.

(65) In some embodiments, the second unit **484** includes a local controller, such as local controller **429** of the first unit **482**. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS devices **421**, **422** and/or a separate heater, if present) in the second unit **484**. In an example, the local controller is integrated into one of the MEMS devices **421**, **422**. In another example, the local controller is coupled to the PCB **430** separate from the MEMS devices **421**, **422**.

(66) In some embodiments, each device in the second unit **484** is independently addressable via a corresponding local controller **429**, such that the operation of each device in the second unit **484** can be controlled without changing the operating status of any other device of showerhead **400A**. In some embodiments, each device in the second unit **484** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers **429**. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **400A** that is not within the defined group.

(67) In a third unit **486**, the PCB **430** includes a first port **432** aligned with a first duct **446** of the manifold **440**, and a second port **434** aligned with a second duct **448** of the manifold **440**. Consequently, the third unit **486** is configured to manage gas supplied via the first conduit **442** of the manifold **440** and via the second conduit **444** of the manifold **440**.

(68) A MEMS device **423** in the void space **474** of the third unit **486** is coupled to the PCB **430**, and regulates gas flow through the first port **432** of the PCB **430** and through the second port **434** of the PCB **430**. It is contemplated that the MEMS device **423** may be configured similarly to MEMS device **250**. MEMS device **423** includes a first orifice **453** and a first valve member **454** for controlling gas flow through the first port **432** of the PCB **430**, and includes a second orifice **455** and a second valve member **456** for controlling gas flow through the second port **434** of the PCB

430. In some embodiments, the MEMS device 423 is soldered to the PCB 430. In some of such embodiments, the solder surrounds the first orifice 453 and/or the second orifice 455, and provides a seal between the PCB 430 and the MEMS device 423. In some embodiments, the MEMS device 423 includes a sensor, such as sensor 236 (FIG. 2B). In some embodiments, a sensor separate from the MEMS device 423 is coupled to the PCB 430. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, the MEMS device 423 includes a heater associated with the first orifice 453. In some embodiments, the MEMS device 423 includes a heater associated with the second orifice 455. In some embodiments, a heater (such as heater 424 of the first unit) separate from the MEMS device 423 is coupled to the PCB 430.

(69) In some embodiments, the third unit 486 includes a local controller, such as local controller 429 of the first unit 482. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device 423 and/or a separate heater, if present) in the third unit 486. In an example, the local controller is integrated into the MEMS device 423. In another example, the local controller is coupled to the PCB 430 separate from the MEMS device 423.

(70) In some embodiments, each device in the third unit 486 is independently addressable via a corresponding local controller 429, such that the operation of each device in the third unit 486 can be controlled without changing the operating status of any other device of showerhead 400A. In some embodiments, each device in the third unit 486 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers 429. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 400A that is not within the defined group.

(71) FIG. 4B is a schematic cross-sectional side view of an exemplary showerhead 400B, which is a variant of showerhead 400A. It is contemplated that the configuration of showerhead 400B may be used as showerhead 112 of FIG. 1.

(72) FIG. 4B depicts the same components as in FIG. 4A, and the above description for FIG. 4A applies for FIG. 4B except that the spacers 460 are omitted, the PCB 430 is attached directly to the top surface 412 of the faceplate 410, and the void spaces 474 are within the faceplate 410. An interface between the faceplate 410 and the PCB 430 is sealed, such as by bonding the faceplate 410 to the PCB 430. As illustrated, in some embodiments, the manifold 440 and PCB 430 are coupled to the faceplate 410 by one or more fastener 416, such as a screw or a bolt. In some embodiments, the fastener 416 may be omitted.

(73) First unit 492, second unit 494, and third unit 496 correspond to the first unit 482, second unit 484, and third unit 486, respectively, of FIG. 4A. As illustrated, in some embodiments, a diffuser 470 is disposed in each compartment 414 above the openings 418 in the faceplate 410. In some embodiments, the diffuser 470 may be omitted.

(74) It is contemplated that clusters of MEMS devices of showerhead 400B may be controlled in similar ways to the examples provided above with respect to clusters of MEMS devices 320 of showerhead 300.

(75) As illustrated, in some embodiments, the first unit 492 includes local controller 429, such as described above. In such embodiments, it is contemplated that the local controller 429 controls operation of at least one device (such as the MEMS device 420 and/or a separate heater 424, if present) in the first unit 492. In an example, the local controller 429 is integrated into the MEMS device 420. In some embodiments, the local controller 429 is coupled to the PCB 430 separate from the MEMS device 420. To inhibit corrosion, it is contemplated that surfaces of the local controller 429 may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as  $\text{Al}_2\text{O}_3$ ), yttrium oxides (such as  $\text{Y}_2\text{O}_3$ ), silicon oxides (such as  $\text{SiO}_x$ ), titanium oxides (such as  $\text{TiO}_2$ ), and the

like.

(76) In some embodiments, each device in the first unit **492** is independently addressable via a corresponding local controller **429**, such that the operation of each device in the first unit **492** can be controlled without changing the operating status of any other device of showerhead **400B**. In some embodiments, each device in the first unit **492** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers **429**. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **400B** that is not within the defined group.

(77) In some embodiments, the second unit **494** includes a local controller, such as local controller **429** of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS devices **421**, **422** and/or a separate heater, if present) in the second unit **494**. In an example, the local controller is integrated into one of the MEMS devices **421**, **422**. In another example, the local controller is coupled to the PCB **430** separate from the MEMS devices **421**, **422**.

(78) In some embodiments, each device in the second unit **494** is independently addressable via a corresponding local controller **429**, such that the operation of each device in the second unit **494** can be controlled without changing the operating status of any other device of showerhead **400B**. In some embodiments, each device in the second unit **494** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers **429**. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **400B** that is not within the defined group.

(79) In some embodiments, the third unit **496** includes a local controller, such as local controller **429** of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device **423** and/or a separate heater, if present) in the third unit **496**. In an example, the local controller is integrated into the MEMS device **423**. In another example, the local controller is coupled to the PCB **430** separate from the MEMS device **423**.

(80) In some embodiments, each device in the third unit **496** is independently addressable via a corresponding local controller **429**, such that the operation of each device in the third unit **496** can be controlled without changing the operating status of any other device of showerhead **400B**. In some embodiments, each device in the third unit **496** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers **429**. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **400B** that is not within the defined group.

(81) In some embodiments, the second unit **484** of FIG. 4A and the second unit **494** of FIG. 4B may be configured to provide independent control of a first gas flow through the first port **432** of the PCB **430** and a second gas flow through the second port **434** of the PCB **430**. For example, an operator can configure the second unit **484** of FIG. 4A or the second unit **494** of FIG. 4B to operate under any one of four modes: (i) flow gas from the first conduit **442** of the manifold **440** only; (ii) flow gas from the second conduit **444** of the manifold **440** only; (iii) flow gas from both the first conduit **442** and the second conduit **444** of the manifold **440**; and (iv) do not flow gas from the manifold **440**.

(82) In the first mode, valve member **426** of MEMS device **421** is positioned to at least partially uncover orifice **425**, and valve member **428** of MEMS device **422** is positioned to obscure orifice **427**. In the second mode, valve member **426** of MEMS device **421** is positioned to obscure orifice **425**, and valve member **428** of MEMS device **422** is positioned to at least partially uncover orifice **427**. In the third mode, valve member **426** of MEMS device **421** is positioned to at least partially

uncover orifice **425**, and valve member **428** of MEMS device **422** is positioned to at least partially uncover orifice **427**. In the fourth mode, valve member **426** of MEMS device **421** is positioned to obscure orifice **425**, and valve member **428** of MEMS device **422** is positioned to obscure orifice **427**.

(83) It is contemplated that an operator can configure the second unit **484** of FIG. **4A** or the second unit **494** of FIG. **4B** to switch from one of the first, second, third, or fourth modes to another of the first, second, third, or fourth modes. In an example, an operator may control the second unit **484** of FIG. **4A** or the second unit **494** of FIG. **4B** to switch between modes in order to adjust a composition of process gases within a zone of a processing volume (**120**, FIG. **1**) of a processing chamber (**100**, FIG. **1**).

(84) In some embodiments, the third unit **486** of FIG. **4A** and the third unit **496** of FIG. **4B** may be configured to provide independent control of a first gas flow through the first port **432** of the PCB **430** and a second gas flow through the second port **434** of the PCB **430**. For example, an operator can configure the third unit **486** of FIG. **4A** or the third unit **496** of FIG. **4B** to operate under any one of four modes: (i) flow gas from the first conduit **442** of the manifold **440** only; (ii) flow gas from the second conduit **444** of the manifold **440** only; (iii) flow gas from both the first conduit **442** and the second conduit **444** of the manifold **440**; and (iv) do not flow gas from the manifold **440**.

(85) In the first mode, first valve member **454** of MEMS device **423** is positioned to at least partially uncover first orifice **453**, and second valve member **456** of MEMS device **423** is positioned to obscure second orifice **455**. In the second mode, first valve member **454** of MEMS device **423** is positioned to obscure first orifice **453**, and second valve member **456** of MEMS device **423** is positioned to at least partially uncover second orifice **455**. In the third mode, first valve member **454** of MEMS device **423** is positioned to at least partially uncover first orifice **453**, and second valve member **456** of MEMS device **423** is positioned to at least partially uncover second orifice **455**. In the fourth mode, first valve member **454** of MEMS device **423** is positioned to obscure first orifice **453**, and second valve member **456** of MEMS device **423** is positioned to obscure second orifice **455**.

(86) It is contemplated that an operator can configure the third unit **486** of FIG. **4A** or the third unit **496** of FIG. **4B** to switch from one of the first, second, third, or fourth modes to another of the first, second, third, or fourth modes. In an example, an operator may control the third unit **486** of FIG. **4A** or the third unit **496** of FIG. **4B** to switch between modes in order to adjust a composition of process gases within a zone of a processing volume (**120**, FIG. **1**) of a processing chamber (**100**, FIG. **1**).

(87) The configuration of manifold **440** and PCB **430** depicted in FIGS. **4A** and **4B** facilitates expedient switching of gases that are flowed through the showerheads **400A** and **400B**. In an example, a first processing operation involves flowing a first gas through first conduit **442**, first ducts **446**, first ports **432**, one or more MEMS devices **420**, **421**, **423**, and through openings **418** of the faceplate **410** into a processing volume of a processing chamber. A second processing operation involves flowing a different second gas through first conduit **442**, first ducts **446**, first ports **432**, one or more MEMS devices **420**, **421**, **423**, and through openings **418** of the faceplate **410** into the processing volume of the processing chamber. In the event that it is detrimental or undesirable to mix the first gas and the second gas, then the first gas must be purged out of the first conduit **442**, the first ducts **446**, the first ports **432**, the one or more MEMS devices **420**, **421**, **423**, and the processing chamber before flowing the second gas. Because the total volume of the first conduit **442** and the first ducts **446** of the manifold plus the first ports **432** of the PCB **430** is less than the volume of a plenum, such as plenum **116**, the quantity of first gas that must be purged and potentially wasted is less for a processing chamber incorporating showerhead **400A** or **400B** than for a processing chamber that delivers process gases via a plenum. Additionally, the time required for the purging operation is less for a processing chamber incorporating showerhead **400A** or **400B** than for a processing chamber that delivers process gases via a plenum. Consequently, a processing



chamber incorporating showerhead **400A** or **400B** provides operational efficiencies of time, gas wastage, throughput, and cost over a processing chamber that delivers process gases via a plenum. (88) The configuration of manifold **440** and PCB **430** depicted in FIGS. **4A** and **4B** facilitates expedient simultaneous delivery of gases that are flowed through the showerheads **400A** and **400B**. In an example, a first gas is flowed through first conduit **442**, first ducts **446**, and one or more MEMS devices **421**, **423**. A second gas is flowed through second conduit **444**, second ducts **448**, and one or more MEMS devices **422**, **423**. The first and second gases are mixed in the void spaces **474** of the second units **484**, **494** and third units **486**, **496** before passing through the openings **418** of faceplate **410**. The MEMS devices **421**, **422**, **423** facilitate localized adjustments to the ratio of first and second gases within each unit **484**, **486**, **494**, **496**. Furthermore, the MEMS devices **421**, **422**, **423** facilitate localized adjustments to the combined flowrate of first and second gases within each unit **484**, **486**, **494**, **496**.

(89) Consequently, the flowrates and relative quantities of the first and second gas within different regions of the processing volume of a processing chamber can be tailored. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted towards being uniform across the processing volume. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted to provide a greater flow of gas in a first region of the processing volume and a lesser flow of gas in a second region of the processing volume. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted to provide a greater proportion of the first gas than the second gas in a first region of the processing volume, and a greater proportion of the second gas than the first gas in a second region of the processing volume.

(90) FIG. **5A** is a schematic cross-sectional side view of an exemplary showerhead **500A**. It is contemplated that the configuration of showerhead **500A** may be used as showerhead **112** of FIG. **1**. Showerhead **500A** includes a faceplate **510A** formed of multiple MEMS modules **520** that are suspended from a PCB **530A**. It is contemplated that the PCB **530A** may be configured similarly to PCB **330** of FIG. **3**. As illustrated, in some embodiments, a portion of the PCB **530A** may extend through the chamber body **102** of the processing chamber. Such a configuration facilitates connection to a master controller **550** external to the processing chamber.

(91) Each MEMS module **520** includes a body **521** with sidewalls **522** containing a MEMS device **560**. In some embodiments, the body **521** is integrated with MEMS device **560**. In some embodiments, the body **521** may be attached to MEMS device **560**. In another embodiment, the body **521** is attached to the PCB **530A** separately from the MEMS device **560**. The sidewalls **522** extend below MEMS device **560** to a base **524**. One or more holes **526** in the base **524** facilitate gases to flow through the MEMS module **560** and into the processing volume (**120**, FIG. **1**) of a processing chamber, such as processing chamber **100**. In some embodiments, a diffuser **528** is disposed above the one or more holes. The diffuser **528** facilitates a uniform distribution of gas through the one or more holes **526**. In some embodiments, the diffuser **528** filters out particles entrained in the gas. Example diffusers **528** include a mesh (such as a sintered mesh), a porous metal filter, a foam (such as porous PTFE foam), or the like. In some embodiments, the diffuser **528** may be omitted.

(92) To inhibit corrosion, it is contemplated that each MEMS module **520** may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal, such as titanium. Additionally, or alternatively, surfaces of each MEMS module **520** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al.sub.2O.sub.3), yttrium oxides (such as Y.sub.2O.sub.3), silicon oxides (such as SiO.sub.x), titanium oxides (such as TiO.sub.2), and the like.

(93) Each MEMS device **560** associated with a corresponding MEMS module **520** may be configured similarly to MEMS device **200**. Each MEMS device **560** is schematically depicted to

include an orifice **572**, a valve member **574**, a heater **576**, and a sensor **578**, such as described above for MEMS device **200**. Each MEMS device **560** is coupled to the PCB **530A**. In some embodiments, the MEMS device **560** is soldered to the PCB **530A**. In some of such embodiments, the solder surrounds the orifice **572** and provides a seal between the PCB **530A** and the MEMS device **560**. Each contact of each MEMS device **560** is connected to the PCB **530A**. The sensor **578**, the heater **576**, and the valve member **574** of each MEMS device **560** receive electrical power via the PCB **530A**. In some embodiments, sensor **578** is omitted, and a sensor separate from the MEMS device **560** is coupled to the PCB **530A**. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, heater **576** is omitted, and a heater separate from the MEMS device **560** is coupled to the PCB **530A**. The PCB **530A** is coupled to a master controller **550** for the transmission of power and/or control signals and/or telemetry with each MEMS device **560**.

(94) The PCB **530A** includes a port **531** associated with each MEMS device **560**. When the valve member **574** of a MEMS device **560** permits gas to flow through the corresponding orifice **572**, gas in the plenum **116** can flow through the corresponding port **531** in the PCB **530A** and through the orifice **572** into the corresponding MEMS module **520**. In some embodiments, the gas is heated by the heater **576**. The gas then flows through the one or more holes **526** in the base **524** of the MEMS module **520**.

(95) In some embodiments, the MEMS module **520** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device **560** and/or a separate heater, if present) in the MEMS module **520**. In an example, the local controller is integrated into the MEMS module **520**, such as by being integrated into the MEMS device **560**. In another example, the local controller is coupled to the PCB **530A** separate from the MEMS device **520**.

(96) In some embodiments, each device (such as MEMS device **560** and/or a separate heater) in the MEMS module **520** is independently addressable via a corresponding local controller, such as local controller **329**, **429**. The operation of each device in the MEMS module **520** can be controlled without changing the operating status of any other device of showerhead **500A**. In some embodiments, each device in the MEMS module **520** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **500A** that is not within the defined group.

(97) It is contemplated that clusters of devices and/or clusters of MEMS modules **520** of showerhead **500A** may be controlled in similar ways to the examples provided above with respect to clusters of MEMS devices **320** of showerhead **300**.

(98) FIG. **5B** is a schematic cross-sectional side view of an exemplary showerhead **500B**, which is a variant of showerhead **500A**, and incorporates elements of showerhead **400B**. It is contemplated that the configuration of showerhead **500B** may be used as showerhead **112** of FIG. **1**. Showerhead **500B** includes a faceplate **510B** formed of multiple MEMS modules **520**, **520A**, **520B** that are suspended from a PCB **530B**. It is contemplated that the PCB **530B** may be configured similarly to PCB **430** of FIGS. **4A** and **4B**. Showerhead **500B** includes a manifold **540** disposed on the PCB **530B**. The manifold **540** includes a first conduit **542** for passage of a first gas, and a second conduit **544** for passage of a second gas. It is contemplated that the first conduit **542** is isolated from the second conduit **544** such that the first gas and the second gas do not mix in the manifold **540**. First ducts **546** from the first conduit **542** are aligned with first ports **532** in the PCB **530B**. Second ducts **548** from the second conduit **544** are aligned with second ports **534** in the PCB **530B**. An interface **538** between the manifold **540** and the PCB **530B** is sealed, such as by bonding the manifold **540** to the PCB **530B**, to inhibit mixing of the first gas and the second gas at the interface **538**. The PCB **530B** is coupled to master controller **550** for the transmission of power and/or control signals

and/or telemetry with MEMS devices attached to the PCB 530B. As illustrated, in some embodiments, a portion of the PCB 530B may extend through the chamber body 102 of the processing chamber. Such a configuration facilitates connection to master controller 550.

(99) In some embodiments, manifold 540 includes one or more additional conduits and corresponding ducts configured to convey one or more additional gases. In such embodiments, the one or more additional conduits may be isolated from the first conduit 542 and the second conduit 544. Furthermore, it is contemplated that the PCB 530B may include additional ports aligned with the additional ducts. As illustrated, in some embodiments, a portion of the manifold 540 may extend through the chamber body 102 of the processing chamber. Such a configuration facilitates connection to one or more gas supplies.

(100) It is contemplated that clusters of devices and/or clusters of MEMS modules 520, 520A, 520B of showerhead 500B may be controlled in similar ways to the examples provided above with respect to clusters of MEMS devices 320 of showerhead 300.

(101) FIG. 5B illustrates three exemplary configurations in which one or more MEMS modules 520, 520A, 520B are attached to the PCB 530B. Each attachment of a MEMS module 520, 520A, 520B with a corresponding portion of the PCB 530B may be considered as a discrete unit; each unit is described below.

(102) A first unit 582 includes MEMS module 520 (as described above), a second unit 584 includes MEMS module 520A, and a third unit 586 includes MEMS module 520B. MEMS module 520A includes a body 521A, sidewalls 522A, and a base 524A with one or more holes 526A, as described above for MEMS module 520. MEMS module 520B includes a body 521B, sidewalls 522B, and a base 524B with one or more holes 526B, as described above for MEMS module 520.

(103) To inhibit corrosion, it is contemplated that each MEMS module 520, 520A, 520B may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal such as titanium. Additionally, or alternatively, surfaces of each MEMS module 520, 520A, 520B may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as  $\text{Al}_2\text{O}_3$ ), yttrium oxides (such as  $\text{Y}_2\text{O}_3$ ), silicon oxides (such as  $\text{SiO}_x$ ), titanium oxides (such as  $\text{TiO}_2$ ), and the like.

(104) In the first unit 582, the PCB 530B includes a port 532 aligned with a first duct 546 of the manifold 540, but does not include a port corresponding to a second duct 548 of the manifold 540. Consequently, the first unit 582 is configured to manage gas supplied via the first conduit 542 of the manifold 540, but is not configured to manage gas supplied via the second conduit 544 of the manifold 540. However, in alternative embodiments, the first unit 582 may be configured to manage gas supplied via the second conduit 544 of the manifold 540, but not configured to manage gas supplied via the first conduit 542 of the manifold 540. In such embodiments, the PCB 530B includes a port aligned with a second duct 548 of the manifold 540, but does not include a port corresponding to a first duct 546 of the manifold 540.

(105) MEMS module 520, as described above, is attached to the PCB 530B. A MEMS device 560 associated with MEMS module 520 coupled to the PCB 530B regulates gas flow through the port 532 of the PCB 530B. It is contemplated that the MEMS device 560 may be configured similarly to MEMS device 200. The MEMS device 560 is schematically depicted to include an orifice 572, a valve member 574, a heater 576, and a sensor 578, such as described above for MEMS device 200. The MEMS device 560 is coupled to the PCB 530B. In some embodiments, the MEMS device 560 is soldered to the PCB 530B. In some of such embodiments, the solder surrounds the orifice 572 and provides a seal between the PCB 530B and the MEMS device 560. Each contact of the MEMS device 560 is connected to the PCB. The sensor 578, the heater 576, and the valve member 574 of the MEMS device 560 receive electrical power via the PCB 530B.

(106) In some embodiments, sensor 578 is omitted, and a sensor separate from the MEMS device 560 is coupled to the PCB 530B. Such a sensor may measure any one or more of pressure,

temperature, or flow rate. In some embodiments, heater **576** is omitted, and a heater separate from the MEMS device **560** is coupled to the PCB **530B**. The PCB **530B** is coupled to a master controller **550** for the transmission of power and/or control signals and/or telemetry with the MEMS device **560**.

(107) In some embodiments, the MEMS module **520** includes a diffuser, such as diffuser **528** described above. In some embodiments, the diffuser may be omitted.

(108) In some embodiments, the first unit **582** includes a local controller, such as local controller **429** of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device **560** and/or a separate heater, if present) in the first unit **582**. In an example, the local controller is integrated into the MEMS module **520**, such as by being integrated into the MEMS device **560**. In another example, the local controller is coupled to the PCB **530B** separate from the MEMS device **560**.

(109) In some embodiments, each device in the first unit **582** is independently addressable via a corresponding local controller (such as local controller **429**), such that the operation of each device in the first unit **582** can be controlled without changing the operating status of any other device of showerhead **500B**. In some embodiments, each device in the first unit **582** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **500B** that is not within the defined group.

(110) In the second unit **584**, the PCB **530B** includes a first port **532** aligned with a first duct **546** of the manifold **540**, and a second port **534** aligned with a second duct **544** of the manifold **540**. Consequently, the second unit **584** is configured to manage gas supplied via the first conduit **542** of the manifold **540** and via the second conduit **544** of the manifold **540**.

(111) MEMS module **520A** is attached to the PCB **530B**. MEMS module **520A** includes a first MEMS device **562** coupled to the PCB **530B** that regulates gas flow through the first port **532** of the PCB **530B**. MEMS module **520A** includes a second MEMS device **564** coupled to the PCB **530B** that regulates gas flow through the second port **534** of the PCB **530B**. It is contemplated that each MEMS device **562**, **564** may be configured similarly to MEMS device **200**.

(112) Each MEMS device **562**, **564** is schematically depicted to include, respectively, an orifice **572A**, **572B**; a valve member **574A**, **574B**; and a heater **576A**, **576B**; such as described above for MEMS device **200**. In some embodiments, heaters **576A**, **576B** may be omitted, and a heater separate from each MEMS device **562**, **564** is coupled to the PCB **530B**. As illustrated, in some embodiments, at least one of the MEMS devices **562**, **564** includes a sensor **578A**, such as sensor **236**. In some embodiments, sensor **578A** is omitted, and a sensor separate from the MEMS devices **562**, **564** is coupled to the PCB **530B**. Such a sensor may measure any one or more of pressure, temperature, or flow rate. Each MEMS device **562**, **564** is coupled to the PCB **530B**. In some embodiments, each MEMS device **562**, **564** is soldered to the PCB **530B**. In some of such embodiments, the solder surrounds each orifice **572A**, **572B** and provides a seal between the PCB **530B** and each MEMS device **562**, **564**. Each contact of each MEMS device **562**, **564** is connected to the PCB **530B**. The sensor **578A**; the heater **576A**, **576B** (including a separate heater when present); and the valve member **574A**, **574B** of each MEMS device **562**, **564** receive electrical power via the PCB **530B**.

(113) In some embodiments, the MEMS module **520A** includes a diffuser, such as diffuser **528** described above. In some embodiments, the diffuser may be omitted.

(114) In some embodiments, the second unit **584** includes a local controller, such as local controller **429** of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS devices **562**, **564** and/or a separate heater, if present) in the second unit **584**. In an example, the local controller is integrated into one of the MEMS devices **562**, **564**. In another example, the local controller is coupled to the PCB **530B** separate from the

MEMS devices **562**, **564**.

(115) In some embodiments, each device in the second unit **584** is independently addressable via a corresponding local controller (such as local controller **429**), such that the operation of each device in the second unit **584** can be controlled without changing the operating status of any other device of showerhead **500B**. In some embodiments, each device in the second unit **584** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **500B** that is not within the defined group.

(116) In some embodiments, the second unit **584** may be configured to provide independent control of a first gas flow through the first port **532** of the PCB **530B** and a second gas flow through the second port **534** of the PCB **530B**. For example, an operator can configure the second unit **584** to operate under any one of four modes: (i) flow gas from the first conduit **542** of the manifold **540** only; (ii) flow gas from the second conduit **544** of the manifold **540** only; (iii) flow gas from both the first conduit **542** and the second conduit **544** of the manifold **540**; and (iv) do not flow gas from the manifold **540**.

(117) In the first mode, valve member **574A** of MEMS device **562** is positioned to at least partially uncover orifice **572A**, and valve member **574B** of MEMS device **564** is positioned to obscure orifice **572B**. In the second mode, valve member **574A** of MEMS device **562** is positioned to obscure orifice **572A**, and valve member **574B** of MEMS device **564** is positioned to at least partially uncover orifice **572B**. In the third mode, valve member **574A** of MEMS device **562** is positioned to at least partially uncover orifice **572A**, and valve member **574B** of MEMS device **564** is positioned to at least partially uncover orifice **572B**. In the fourth mode, **574A** of MEMS device **562** is positioned to obscure orifice **572A**, and valve member **574B** of MEMS device **564** is positioned to obscure orifice **572B**.

(118) It is contemplated that an operator can configure the second unit **584** to switch from one of the first, second, third, or fourth modes to another of the first, second, third, or fourth modes. In an example, an operator may control the second unit **584** to switch between modes in order to adjust a composition of process gases within a zone of a processing volume (**120**, FIG. **1**) of a processing chamber (**100**, FIG. **1**).

(119) In the third unit **586**, the PCB **530B** includes a first port **532** aligned with a first duct **546** of the manifold **540**, and a second port **534** aligned with a second duct **548** of the manifold **540**. Consequently, the third unit **586** is configured to manage gas supplied via the first conduit **542** of the manifold **540** and via the second conduit **542** of the manifold **540**.

(120) MEMS module **520B** is attached to the PCB **530B**. MEMS module **520B** includes a MEMS device **566** coupled to the PCB **530B** that regulates gas flow through the first port **532** of the PCB **530B** and regulates gas flow through the second port **534** of the PCB **530B**. It is contemplated that the MEMS device **566** may be configured similarly to MEMS device **250**.

(121) The MEMS device **566** is schematically depicted to include a first orifice **572C** and a first valve member **574C** for controlling gas flow through the first port **532** of the PCB **530B**, such as described above for MEMS device **250**. The MEMS device **566** is schematically depicted to include a second orifice **572D** and a second valve member **574D** for controlling gas flow through the second port **534** of the PCB **530B**, such as described above for MEMS device **250**. In some embodiments, the MEMS device **566** is soldered to the PCB **530B**. In some of such embodiments, the solder surrounds the first orifice **572C** and the second orifice **572D**, and provides a seal between the PCB **530B** and the MEMS device **566**. The MEMS device **566** is schematically depicted to include a first heater **576C** associated with the first orifice **572C** and a second heater **576D** associated with the second orifice **572D**, such as described above for MEMS device **250**. In some embodiments, heaters **576C**, **576D** are omitted, and a heater separate from the MEMS device **566** is coupled to the PCB **530B**. The MEMS device **566** is schematically depicted to include a

sensor **578B**, such as described above for MEMS device **250**. In some embodiments, sensor **578B** is omitted, and a sensor separate from the MEMS device **566** is coupled to the PCB **530B**. Such a sensor may measure any one or more of pressure, temperature, or flow rate. The sensor **578B**, the heaters **576C**, **576D**, and the valve members **574C**, **574D** of the MEMS device **566**, and/or a separate heater (if present) receive electrical power via the PCB **530B**.

(122) In some embodiments, the MEMS module **520B** includes a diffuser, such as diffuser **528** described above. In some embodiments, the diffuser may be omitted.

(123) In some embodiments, the third unit **586** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device **566** and/or a separate heater, if present) in the third unit **586**. In an example, the local controller is integrated into the MEMS device **566**. In another example, the local controller is coupled to the PCB **530B** separate from the MEMS device **566**.

(124) In some embodiments, each device in the third unit **586** is independently addressable via a corresponding local controller (such as local controller **429**), such that the operation of each device in the third unit **586** can be controlled without changing the operating status of any other device of showerhead **500B**. In some embodiments, each device in the third unit **586** is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead **500B** that is not within the defined group.

(125) In some embodiments, the third unit **586** may be configured to provide independent control of a first gas flow through the first port **532** of the PCB **530B** and a second gas flow through the second port **534** of the PCB **530B**. For example, an operator can configure the third unit **586** to operate under any one of four modes: (i) flow gas from the first conduit **542** of the manifold **540** only; (ii) flow gas from the second conduit **544** of the manifold **540** only; (iii) flow gas from both the first conduit **542** and the second conduit **544** of the manifold **540**; and (iv) do not flow gas from the manifold **540**.

(126) In the first mode, first valve member **574C** of MEMS device **566** is positioned to at least partially uncover first orifice **572C**, and second valve member **574D** of MEMS device **566** is positioned to obscure second orifice **572D**. In the second mode, first valve member **574C** of MEMS device **566** is positioned to obscure first orifice **572C**, and second valve member **574D** of MEMS device **566** is positioned to at least partially uncover second orifice **572D**. In the third mode, first valve member **574C** of MEMS device **566** is positioned to at least partially uncover first orifice **572C**, and second valve member **574D** of MEMS device **566** is positioned to at least partially uncover second orifice **572D**. In the fourth mode, first valve member **574C** of MEMS device **566** is positioned to obscure first orifice **572C**, and second valve member **574D** of MEMS device **566** is positioned to obscure second orifice **572D**.

(127) It is contemplated that an operator can configure the third unit **586** to switch from one of the first, second, third, or fourth modes to another of the first, second, third, or fourth modes. In an example, an operator may control the third unit **586** to switch between modes in order to adjust a composition of process gases within a zone of a processing volume (**120**, FIG. **1**) of a processing chamber (**100**, FIG. **1**).

(128) The configuration of manifold **540** and PCB **530B** depicted in FIG. **5B** facilitates expedient switching of gases that are flowed through showerhead **500B**. In an example, a first processing operation involves flowing a first gas through first conduit **542**, first ducts **546**, first ports **532**, one or more MEMS devices **560**, **562**, **566**, and through the one or more holes **526**, **526A**, **526B** of the faceplate **510B** into a processing volume of a processing chamber. A second processing operation involves flowing a different second gas through first conduit **542**, first ducts **546**, first ports **532**, one or more MEMS devices **560**, **562**, **566**, and through the one or more holes **526**, **526A**, **526B** of

the faceplate **510B** into the processing volume of the processing chamber. In the event that it is detrimental or undesirable to mix the first gas and the second gas, then the first gas must be purged out of the first conduit **542**, the first ducts **546**, the first ports **532**, the one or more MEMS devices **560**, **562**, **566**, and the processing chamber before flowing the second gas. Because the total volume of the first conduit **442** and the first ducts **446** of the manifold plus the first ports **532** of the PCB **530B** is less than the volume of a plenum, such as plenum **116**, the quantity of first gas that must be purged and potentially wasted is less for a processing chamber incorporating showerhead **500 B** than for a processing chamber that delivers process gases via a plenum. Additionally, the time required for the purging operation is less for a processing chamber incorporating showerhead **500 B** than for a processing chamber that delivers process gases via a plenum. Consequently, a processing chamber incorporating showerhead **500 B** provides operational efficiencies of time, gas wastage, throughput, and cost over a processing chamber that delivers process gases via a plenum. (129) The configuration of manifold **540** and PCB **530B** depicted in FIG. **5B** facilitates expedient simultaneous delivery of gases that are flowed through showerhead **500B**. In an example, a first gas is flowed through first conduit **542**, first ducts **546**, first ports **532**, and one or more MEMS devices **562**, **566**. A second gas is flowed through second conduit **544**, second ducts **548**, second ports **534**, and one or more MEMS devices **564**, **566**. The first and second gases are mixed in the MEMS modules **520A**, **520B** of the second **584** and third **586** units, respectively, before passing through the one or more holes **526A**, **526B**, respectively, of the faceplate **510B**. The MEMS devices **562**, **564**, **566** facilitate localized adjustments to the ratio of first and second gases within each unit **584**, **586**. Furthermore, the MEMS devices **562**, **564**, **566** facilitate localized adjustments to the combined flowrate of first and second gases within each unit **584**, **586**.

(130) Consequently, the flowrates and relative quantities of the first and second gas within different regions of the processing volume of a processing chamber can be tailored. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted towards being uniform across the processing volume. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted to provide a greater flow of gas in a first region of the processing volume and a lesser flow of gas in a second region of the processing volume. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted to provide a greater proportion of the first gas than the second gas in a first region of the processing volume, and a greater proportion of the second gas than the first gas in a second region of the processing volume.

(131) FIGS. **6A-6G** are schematic cross-sectional side views of different configurations of MEMS modules. In some embodiments, it is contemplated that showerhead **500A** and showerhead **500B** may include MEMS modules of the same configuration. Additionally, or alternatively, it is contemplated that showerhead **500A** and showerhead **500B** may include MEMS modules of different configurations.

(132) FIG. **6A** illustrates MEMS module **600A**. It is contemplated that MEMS module **600A** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600A** is shown as including MEMS device **610A**. It is contemplated that MEMS device **610A** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610A** is schematically depicted to include an orifice **612A**, a valve member **614A**, a heater **616A** at the orifice **612A**, and a sensor **618A**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600A** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600A** may contain more than one MEMS device.

(133) MEMS module **600A** includes a body **601A** with sidewalls **602A**. In some embodiments, the body **601A** is integrated with MEMS device **610A**. In some embodiments, the body **601A** may be attached to MEMS device **610A**. In another embodiment, the body **601A** is configured to be attached to a PCB separately from MEMS device **610A**. The sidewalls **602A** extend below MEMS

device **610A** to a base **604A**. One or more holes **606A** in the base **604A** facilitate gases to flow through the MEMS device **610A**. The base **604A** is configured as an insert that is coupled to the sidewalls **602A**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602A** by a screw thread. In some embodiments, a diffuser, such as diffuser **470**, is disposed above the one or more holes **606A**.

(134) In some embodiments, MEMS module **600A** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610A**. In an example, the local controller is integrated into the MEMS device **610A**. In another example, the local controller is coupled to a PCB separately from MEMS device **610A**.

(135) FIG. **6B** illustrates MEMS module **600B**. It is contemplated that MEMS module **600B** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600B** is shown as including MEMS device **610B**. It is contemplated that MEMS device **610B** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610B** is schematically depicted to include an orifice **612B**, a valve member **614B**, a heater **616B** at the orifice **612B**, and a sensor **618B**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600B** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600B** may contain more than one MEMS device.

(136) MEMS module **600B** includes a body **601B**. In some embodiments, the body **601B** is integrated with MEMS device **610B**. In some embodiments, the body **601B** may be attached to MEMS device **610B**. In another embodiment, the body **601B** is configured to be attached to a PCB separately from MEMS device **610B**. An insert holder **607** is attached to the body **601B**, such as by a weld or an adhesive bond. The insert holder **607** includes sidewalls **602B** extending below MEMS device **610B** to a base **604B**. One or more holes **606B** in the base **604B** facilitate gases to flow through the MEMS device **610B**. The base **604B** is configured as an insert that is coupled to the sidewalls **602B**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602B** by a screw thread. In some embodiments, a diffuser, such as diffuser **470**, is disposed above the one or more holes **606B**.

(137) In some embodiments, MEMS module **600B** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610B**. In an example, the local controller is integrated into the MEMS device **610B**. In another example, the local controller is coupled to a PCB separately from MEMS device **610B**.

(138) FIG. **6C** illustrates MEMS module **600C**. It is contemplated that MEMS module **600C** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600C** is shown as including MEMS device **610C**. It is contemplated that MEMS device **610C** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610C** is schematically depicted to include an orifice **612C**, a valve member **614C**, a heater **616C** at the orifice **612C**, and a sensor **618C**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600C** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600C** may contain more than one MEMS device.

(139) MEMS module **600C** includes a body **601C**. In some embodiments, the body **601C** is



integrated with MEMS device **610C**. In some embodiments, the body **601C** may be attached to MEMS device **610C**. In another embodiment, the body **601C** is configured to be attached to a PCB separately from MEMS device **610C**. A base unit **608** is attached to the body **601C**, such as by a weld or an adhesive bond. The base unit **608** includes sidewalls **602C** extending below MEMS device **610C** to a base **604C**. One or more holes **606C** in the base **604C** facilitate gases to flow through the MEMS device **610C**. The base **604B** is affixed to, or is integral with, the sidewalls **602C**. In some embodiments, a diffuser, such as diffuser **470**, is disposed above the one or more holes **606C**.

(140) In some embodiments, MEMS module **600C** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610C**. In an example, the local controller is integrated into the MEMS device **610C**. In another example, the local controller is coupled to a PCB separately from MEMS device **610C**.

(141) FIG. **6D** illustrates MEMS module **600D**. It is contemplated that MEMS module **600D** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600D** is shown as including MEMS device **610D**. It is contemplated that MEMS device **610D** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610D** is schematically depicted to include an orifice **612D**, a valve member **614D**, and a sensor **618D**, such as described above for MEMS device **200**. A heater **622** is attached to MEMS device **610D**, and extends below MEMS device **610D**. In some embodiments, the heater **622** includes a mesh, such as a sintered mesh. In some embodiments, the heater **622** also serves as a diffuser or a filter. In some embodiments, it is contemplated that MEMS module **600D** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600D** may contain more than one MEMS device.

(142) MEMS module **600D** includes a body **601D** with sidewalls **602D**. In some embodiments, the body **601D** is integrated with MEMS device **610D**. In some embodiments, the body **601D** may be attached to MEMS device **610D**. In another embodiment, the body **601D** is configured to be attached to a PCB separately from MEMS device **610D**. The sidewalls **602D** extend below MEMS device **610D** to a base **604D**. One or more holes **606D** in the base **604D** facilitate gases to flow through the MEMS device **610D**. The base **604D** is configured as an insert that is coupled to the sidewalls **602D**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602D** by a screw thread. In some embodiments, a diffuser, such as diffuser **470**, is disposed above the one or more holes **606D**.

(143) In some embodiments, MEMS module **600D** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610D**. In an example, the local controller is integrated into the MEMS device **610D**. In another example, the local controller is coupled to a PCB separately from MEMS device **610D**.

(144) FIG. **6E** illustrates MEMS module **600E**. It is contemplated that MEMS module **600E** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600E** is shown as including MEMS device **610E**. It is contemplated that MEMS device **610E** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610E** is schematically depicted to include an orifice **612E**, a valve member **614E**, and a sensor **618E**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600E** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600E** may contain more than one MEMS device.

(145) MEMS module **600E** includes a body **601E** with sidewalls **602E**. In some embodiments, the

body **601E** is integrated with MEMS device **610E**. In some embodiments, the body **601E** may be attached to MEMS device **610E**. In another embodiment, the body **601E** is configured to be attached to a PCB separately from MEMS device **610E**. The sidewalls **602E** extend below MEMS device **610E** to a base **604E**. One or more holes **606E** in the base **604E** facilitate gases to flow through the MEMS device **610E**. The base **604E** is configured as an insert that is coupled to the sidewalls **602E**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602E** by a screw thread. In some embodiments, a diffuser, such as diffuser **470**, is disposed above the one or more holes **606E**.

(146) MEMS module **600E** includes a heater **624** that is separate from MEMS device **610E**. Heater **624** is attached to body **601E**. It is contemplated that electrical connections between the heater **624** and a PCB may be facilitated by wiring through the body **601E** and/or through MEMS device **610E**. As illustrated, it is contemplated that heater **624** may be located below MEMS device **610E**. In some embodiments, the heater **624** includes a mesh, such as a sintered mesh. In some embodiments, the heater **624** also serves as a diffuser or a filter.

(147) In some embodiments, MEMS module **600E** includes a local controller, such as local controller **429** of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610E**. In an example, the local controller is integrated into the MEMS device **610E**. In another example, the local controller is coupled to a PCB separately from MEMS device **610E**.

(148) FIG. 6F illustrates MEMS module **600F**. It is contemplated that MEMS module **600F** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600F** is shown as including MEMS device **610F**. It is contemplated that MEMS device **610F** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610F** is schematically depicted to include an orifice **612F**, a valve member **614F**, and a sensor **618F**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600F** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600F** may contain more than one MEMS device.

(149) MEMS module **600F** includes a body **601F** with sidewalls **602F**. In some embodiments, the body **601F** is integrated with MEMS device **610F**. In some embodiments, the body **601F** may be attached to MEMS device **610F**. In another embodiment, the body **601F** is configured to be attached to a PCB separately from MEMS device **610F**. The sidewalls **602F** extend below MEMS device **610F** to a base **604F**. One or more holes **606F** in the base **604F** facilitate gases to flow through the MEMS device **610F**. The base **604F** is configured as an insert that is coupled to the sidewalls **602F**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602F** by a screw thread.

(150) MEMS module **600F** includes a heater **626** that is separate from MEMS device **610F**. Heater **626** is attached to body **601F**. It is contemplated that electrical connections between the heater **626** and a PCB may be facilitated by wiring through the body **601F** and/or through MEMS device **610F**. In some embodiments, the heater **626** includes a mesh, such as a sintered mesh. In some embodiments, the heater **626** also serves as a diffuser or a filter.

(151) As illustrated, it is contemplated that heater **626** may be located below MEMS device **610F** at, or close to, the base **604F**. As illustrated, in some embodiments it is contemplated that a diffuser **628**, such as diffuser **470**, may be located between the heater **626** and the MEMS device **610F**.

(152) In some embodiments, MEMS module **600F** includes a local controller, such as local

controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610F**. In an example, the local controller is integrated into the MEMS device **610F**. In another example, the local controller is coupled to a PCB separately from MEMS device **610F**.

(153) FIG. **6G** illustrates MEMS module **600G**. It is contemplated that MEMS module **600G** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600G** is shown as including MEMS device **610G**. It is contemplated that MEMS device **610G** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610G** is schematically depicted to include an orifice **612G**, a valve member **614G**, and a sensor **618G**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600G** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600G** may contain more than one MEMS device.

(154) MEMS module **600G** includes a body **601G** with sidewalls **602G**. In some embodiments, the body **601G** is integrated with MEMS device **610G**. In some embodiments, the body **601G** may be attached to MEMS device **610G**. In another embodiment, the body **601G** is configured to be attached to a PCB separately from MEMS device **610G**. The sidewalls **602G** extend below MEMS device **610G** to a base **604G**. One or more holes **606G** in the base **604G** facilitate gases to flow through the MEMS device **610G**. The base **604G** is configured as an insert that is coupled to the sidewalls **602G**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602G** by a screw thread. As illustrated, in some embodiments, a diffuser **628**, such as diffuser **470**, is disposed above the one or more holes **606G**.

(155) It is contemplated that at least the base **604G** may serve as a heating element. As an example, the base may be manufactured from graphite, such as in the form of IFS-2B. In some embodiments, the sidewalls **602G** are manufactured from a material similar to that of the base **604G**. In some embodiments, the body **601G** is manufactured from a material similar to that of the base **604G**. It is contemplated that any part of the body **601G**, sidewalls **602G**, or base **604G** that is not manufactured to serve as a heating element may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal such as titanium.

(156) In some embodiments, MEMS module **600G** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610G**. In an example, the local controller is integrated into the MEMS device **610G**. In another example, the local controller is coupled to a PCB separately from MEMS device **610G**.

(157) To inhibit corrosion and/or reduce a probability that a valve member of a MEMS device may stick in position and become inoperable, it is contemplated that surfaces of each MEMS device **610A-610G**, and/or local controller (if present), and/or heater **622**, **624**, **626**, and/or diffuser (if present) may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as  $\text{Al.sub.2O.sub.3}$ ), yttrium oxides (such as  $\text{Y.sub.2O.sub.3}$ ), silicon oxides (such as  $\text{SiO.sub.x}$ ), titanium oxides (such as  $\text{TiO.sub.2}$ ), and the like.

(158) To inhibit corrosion, it is contemplated that each MEMS module **600A-600F** may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal such as titanium. Additionally, or alternatively, surfaces of each MEMS module **600A-600F** and **600G** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as  $\text{Al.sub.2O.sub.3}$ ), yttrium oxides (such as  $\text{Y.sub.2O.sub.3}$ ), silicon oxides (such as  $\text{SiO.sub.x}$ ), titanium oxides (such as  $\text{TiO.sub.2}$ ), and the like.

(159) FIG. 7 is a schematic plan view of a faceplate **700** of a showerhead. It is contemplated that faceplate **700** may represent any faceplate of the present disclosure, such as any of faceplates **310**, **410**, **510A**, **510B** in a corresponding showerhead **300**, **400A**, **400B**, **500A**, **500B** that is used as showerhead **112** of processing chamber **100**. Zones **710** are delineated on faceplate **700**, each zone **710** includes one or more openings **712** through the faceplate **700**. In operation, process gas flow through the one or more openings **712** of a specific zone **710** is controlled by one or more MEMS devices associated with the specific zone **710**. In operation, one or more devices, such as MEMS devices and/or heaters, are associated with each zone **710**, and are controlled independently and/or in groups, as described above. In an example, the operation of each MEMS device or heater can be controlled without changing the operating status of another MEMS device or heater associated with faceplate **700**. It is contemplated that the number, size, and distribution of zones **710** may be configured according to the type of process to be conducted in processing chamber **100** and/or the specific process gas(es) to be employed.

(160) It is contemplated that any of PCB **330**, **430**, **530A**, **530B** may be made from a ceramic material with metal conductors embedded therein. In some embodiments, it is contemplated that PCBs **530A** and **530B** may include a provision for coupling to one or more support members. In an example, the one or more support members facilitate PCBs **530A** and **530B** spanning a processing volume in a processing chamber and bearing the weight of components, such as MEMS modules **520**, **520A**, **520B**, that are suspended from PCBs **530A** and **530B**. In a further example, the manifold **540** acts as a support member for PCB **530B**.

(161) It is contemplated that manifold **440** or **540** may be made from a ceramic material. In some embodiments, it is contemplated that manifold **440** or **540** may include a provision for coupling to one or more support members. In an example, the one or more support members facilitate manifold **440** or **540** spanning a processing volume in a processing chamber and bearing the weight of components, such as PCB **430** or **530B**, that are suspended from manifold **440** or **540**. In a further example, the PCB **530B** acts as a support member for manifold **540**.

(162) It is contemplated that any of master controllers **350**, **450**, **550** includes a central processing unit (CPU), a memory containing instructions, and support circuits for the CPU. The master controller **350**, **450**, **550** is of any form of a general-purpose computer processor that is used in an industrial setting for controlling various chambers and equipment and/or sub-processors thereon or therein.

(163) The memory, or non-transitory computer readable medium, is one or more of a readily available memory such as random access memory (RAM), read only memory (ROM), floppy disk, hard disk, flash drive, or any other form of digital storage, local or remote. The support circuits are coupled to the CPU for supporting the CPU (a processor). The support circuits include cache, power supplies, clock circuits, input/output circuitry and subsystems, and the like. Operations and operating parameters are stored in the memory as a software routine that is executed or invoked to turn the master controller **350**, **450**, **550** into a specific purpose controller to control the operations of any of showerhead **300**, **400A**, **400B**, **500A**, **500B**. The master controller **350**, **450**, **550** is configured to conduct any of the operations described herein. The instructions stored on the memory, when executed, cause one or more of the operations described herein to be conducted.

(164) In some embodiments, data from any of sensors **236**, **328**, **578**, **578A**, **578B**, **618A**, **618B**, **618C**, **618D**, **618E**, **618F**, **618G** and/or any sensor associated with any of showerhead **300**, **400A**, **400B**, **500A**, **500B** may be used to provide feedback to a controller, such as any of master controller **350**, **450**, **550**. In some embodiments, data of electrical current through any of heaters **326**, **424**, **576**, **576A**, **576B**, **576C**, **576D**, **616A**, **616B**, **616C**, **622**, **624**, **626**, and/or any heater associated with any of showerhead **300**, **400A**, **400B**, **500A**, **500B** may be used to provide feedback to a controller, such as any of master controller **350**, **450**, **550**. For example, the data may include pressure data and/or temperature data.

(165) The controller, such as any of master controller **350**, **450**, **550** uses the data so provided as an

input to process commands addressed to individual MEMS device(s) and/or group(s) of MEMS devices. In some of such embodiments, the commands prompt the individual MEMS device(s) and/or group(s) of MEMS devices to adjust a flowrate of a gas through the individual MEMS device(s) and/or group(s) of MEMS devices. In an example, such adjustments provide control of flow distribution of one or more gases through any of showerhead **300, 400A, 400B, 500A, 500B**. In some of such embodiments, the commands prompt the individual MEMS device(s) and/or group(s) of MEMS devices to adjust a temperature of a gas flowing through the individual MEMS device(s) and/or group(s) of MEMS devices. In an example, such adjustments provide control of temperature distribution of one or more gases through any of showerhead **300, 400A, 400B, 500A, 500B**.

(166) The instructions in the memory of the master controller **350, 450, 550** can include one or more machine learning/artificial intelligence algorithms that can be executed in addition to the operations described herein. As an example, a machine learning/artificial intelligence algorithm executed by the master controller **350, 450, 550** can optimize and alter operational parameters based on one or more sensor measurements taken by the one or more sensors **236, 328, 578, 578A, 578B, 618A-618G**. The operational parameters can include, for example, pressure, temperature, gas flow rate, valve member position, and heater status.

(167) The one or more machine learning/artificial intelligence algorithms can account for variations in gas flow rate, gas ratios, temperature, and pressure across any of the showerheads described herein. In some embodiments, the one or more machine learning/artificial intelligence algorithms can measure any of the above parameters to determine whether an appropriate quantity of a process gas at an appropriate pressure and an appropriate temperature is being delivered to the appropriate regions of a processing volume according to a prescribed operation. In some embodiments, the one or more machine learning/artificial intelligence algorithms can prompt the master controller **350, 450, 550** to initiate corrective action in order to adjust the quantity, pressure, or temperature of a process gas being delivered to a prescribed region of a processing volume.

(168) The operational flexibilities described above facilitated by showerheads of the present disclosure are not provided by conventional showerheads. Conventional showerheads typically are configured for achieving an optimal distribution of a specific process gas by having a given number of openings of particular sizes arranged in a specific pattern. However, the number, sizing, and arrangement of openings that provides for an optimal distribution of a one process gas do not necessarily provide for an optimal distribution of a different process gas. In contrast, showerheads of the present disclosure provide for the relative flow rate of any process gas through an opening, or a cluster of openings, to be tailored provide for an optimal distribution of any process gas.

(169) The operational flexibilities facilitated by showerheads of the present disclosure provide time and cost efficiencies compared to conventional operations. For example, some conventional processing sequences involve performing a first operation on a substrate with a first process gas in a first processing chamber, then transferring the substrate to a second processing chamber, then performing a second operation on the substrate with a second, different, process gas. In contrast, processing chambers incorporating showerheads of the present disclosure may be used to perform both the first operation with the first process gas and then the second operation with the second process gas. Thus, the need for the second processing chamber is alleviated, as is the time taken to transfer the substrate between the processing chambers.

(170) In another example, some conventional processing sequences involve performing a first operation on a substrate with a first process gas in a first processing chamber including a first conventional showerhead, then transferring the substrate to a second processing chamber that includes a second conventional showerhead, then performing a second operation on the substrate in the processing chamber with a second, different, process gas. In contrast, processing chambers incorporating showerheads of the present disclosure may be used to perform both the first operation with the first process gas and then the second operation with the second process gas using the same

showerhead in the same processing chamber. Thus, the need for transferring the substrate between processing chambers is alleviated, as is the time taken to transfer the substrate.

(171) Furthermore, the operational flexibilities facilitated by showerheads of the present disclosure provide efficiencies in inventory management compared to conventional operations. The use of showerheads of the present disclosure reduces or eliminates the need to stock different showerheads that are configured for use with different process gases for different process operations.

(172) It is contemplated that elements and features of any one disclosed embodiment may be beneficially incorporated in one or more other embodiments. While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

## Claims

1. A showerhead for a processing chamber, the showerhead comprising: a printed circuit board including a plurality of ports therethrough; and a faceplate including: a plurality of MEMS modules coupled to the printed circuit board, each MEMS module including: a body with sidewalls extending to a base, the base including a plurality of holes; and a MEMS device disposed in the body, the MEMS device including an orifice and a corresponding valve member operable to control fluid flow from at least one of the plurality of ports to the plurality of holes.
2. The showerhead of claim 1, wherein the valve member is movable between first and second positions to regulate the fluid flow through the orifice.
3. The showerhead of claim 2, wherein each MEMS module includes a heater configured to heat a fluid conveyed through the orifice.
4. The showerhead of claim 2, wherein the valve member blocks fluid flow through the orifice when in the first position.
5. The showerhead of claim 1, further comprising a local controller coupled to the printed circuit board, the local controller configured to control operation of a corresponding MEMS device of the plurality of MEMS modules.
6. The showerhead of claim 1, further comprising a manifold coupled to the printed circuit board, the manifold including a first conduit and a plurality of first ducts, each first duct associated with a corresponding first port of the plurality of ports of the printed circuit board.
7. The showerhead of claim 6, wherein the manifold includes a second conduit and a plurality of second ducts, each second duct associated with a corresponding second port of the plurality of ports of the printed circuit board.
8. The showerhead of claim 7, wherein: the MEMS device includes a first MEMS device and a second MEMS device; the first MEMS device is associated with a corresponding first port of the plurality of ports; and the second MEMS device is associated with a corresponding second port of the plurality of ports.
9. The showerhead of claim 1, wherein each MEMS module of the plurality of MEMS modules is suspended from the printed circuit board.
10. The showerhead of claim 1, wherein each MEMS device is suspended from the printed circuit board.
11. The showerhead of claim 1, wherein at least one MEMS module of the plurality of MEMS modules includes a diffuser disposed above the plurality of holes.
12. A showerhead for a processing chamber, the showerhead comprising: a printed circuit board including a plurality of ports therethrough; and a faceplate including: a plurality of MEMS modules suspended from the printed circuit board, each MEMS module including: a body with sidewalls extending to a base, the base including a plurality of holes; and a MEMS device disposed in the

body and spaced from the plurality of holes, the MEMS device including an orifice and a valve member operable to control fluid flow from at least one of the plurality of ports, through the orifice, and to the plurality of holes.

13. The showerhead of claim 12, wherein: the valve member is movable between first and second positions to regulate fluid flow through the orifice; and the valve member blocks fluid flow through the orifice when in the first position.

14. The showerhead of claim 12, wherein at least one MEMS module of the plurality of MEMS modules includes a heater configured to heat a fluid conveyed through the orifice.

15. The showerhead of claim 12, wherein at least one MEMS module of the plurality of MEMS modules includes a diffuser disposed above the plurality of holes.

16. The showerhead of claim 12, further comprising a local controller coupled to the printed circuit board, the local controller configured to control operation of a corresponding MEMS device of the plurality of MEMS modules.

17. The showerhead of claim 12, further comprising a manifold coupled to the printed circuit board, the manifold including a first conduit and a plurality of first ducts, each first duct associated with a corresponding first port of the plurality of ports of the printed circuit board.

18. The showerhead of claim 17, wherein the manifold includes a second conduit and a plurality of second ducts, each second duct associated with a corresponding second port of the plurality of ports of the printed circuit board.

19. The showerhead of claim 18, wherein: the MEMS device includes a first MEMS device and a second MEMS device; the first MEMS device is associated with a corresponding first port of the plurality of ports; and the second MEMS device is associated with a corresponding second port of the plurality of ports.

20. A processing chamber comprising: a chamber body; a printed circuit board including a plurality of ports therethrough; and a showerhead disposed in the chamber body, the showerhead comprising: a faceplate including: a plurality of MEMS modules coupled to the printed circuit board, each MEMS module including: a body with sidewalls extending to a base, the base including a plurality of holes; and a MEMS device disposed in the body, the MEMS device including an orifice and a corresponding valve member operable to control fluid flow from at least one of the plurality of ports to the plurality of holes and a controller coupled to the printed circuit board, and configured to control operations at least one MEMS device of the plurality of MEMS devices independently of operations of other MEMS devices of the plurality of MEMS devices.

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